Spatio-temporal dynamics of soil organic carbon in Ghanaian croplands under climate and land vegetation productivity changes

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Abstract

Agriculture remains central to Ghana's economy, providing food, employment, and income for most rural households. Yet recent intensification through mechanization and high input use has accelerated soil degradation, particularly the decline of soil organic carbon (SOC), a key indicator of soil fertility and climate regulation. This study applied the Rothamsted Carbon Model (RothC) to quantify and project SOC dynamics in Ghanaian croplands from 2001 to 2024 and to simulate trajectories to 2074 under baseline, optimistic, and pessimistic climate and vegetation productivity trends. Harmonized datasets on SOC, clay content, NDVI, and climate variables were processed at 250 m resolution in Google Earth Engine. Model outputs correlated strongly with observed SOC ($r = 0.91$) but yielded a negative Nash–Sutcliffe efficiency (–1.16), reflecting systematic underestimation relative to SoilGrids while remaining consistent with field measurements. National mean SOC declined from 23.79 t ha^{-1} in 2001 to 22.94 t ha^{-1} in 2024 (–3.6%), with the steepest reductions in southern high-SOC zones and slight gains in northern low-SOC areas. Across agroecological zones, losses ranged from 27–31% ($p < 0.0001$), primarily due to humus pool depletion, whereas inert organic matter remained stable. Lag analyses indicated immediate SOC reductions following temperature increases and delayed effects under moisture stress. Projections suggest further SOC declines of 17–35% by 2074, underscoring the need for climate-responsive and agroecologically differentiated soil carbon management strategies in Ghana. Integrating SOC monitoring into national climate and land restoration policies would enable more targeted interventions to sustain soil productivity and carbon resilience.

Keywords

Soil organic carbon, RothC model, climate change, agroecological zones, carbon sequestration, climate-smart agriculture.

1. Introduction

Agriculture remains a cornerstone of Ghana's economy, supporting food supply, employment, and rural livelihoods (Bawa, 2019). Yet the sector faces mounting pressure from land degradation, declining soil fertility, and increasing climate variability. Recent intensification through mechanization and heavy reliance on external inputs has produced mixed outcomes - raising yields in some systems while accelerating soil depletion and nutrient loss in others. These trends threaten the long-term sustainability of Ghana's predominantly smallholder agriculture.

Soil organic carbon (SOC) is a fundamental indicator of soil health, influencing nutrient availability, water retention, and the regulation of atmospheric CO_2 (Lal, 2004). It represents the balance between organic inputs from vegetation and outputs driven by decomposition and disturbance. Even modest declines in SOC stocks can substantially reduce soil productivity and resilience. In sub-Saharan Africa, widespread SOC depletion has been linked to land-use change, residue removal, and limited vegetative cover (Sommer et al., 2019) . Quantifying SOC changes through time therefore provides an essential basis for evaluating soil condition and the effects of climate and vegetation variability on agricultural systems.

Despite Ghana's growing recognition of soil degradation as a national concern, the spatial and temporal behaviour of SOC has not been systematically quantified. Most existing assessments remain site-specific or static, providing little insight into how SOC responds to changing climatic and vegetation conditions across agroecological zones. The increasing availability of spatial datasets, including satellite-derived indicators such as the Normalized Difference Vegetation Index (NDVI), now offers new opportunities to evaluate SOC dynamics over time and at national scale.

This study employs the Rothamsted Carbon Model (RothC) (Coleman & Jenkinson, 1996) to simulate SOC dynamics in Ghanaian croplands from 2001 to 2024 and to project future trajectories to 2074 under alternative climate and vegetation productivity trends. Using harmonized datasets on SOC, clay content, NDVI, temperature, and rainfall at 250 m resolution, the analysis provides the first country-wide assessment of SOC change and its drivers across Ghana's agroecological zones. The results aim to guide the development of climate-responsive, evidence-based soil management policies that sustain fertility, enhance carbon sequestration, and strengthen agricultural resilience.

2. Materials and Methods

2.1. Scope of the study

This study focuses on Ghana, located in West Africa between 4.5°–11.5°N and 3.5°W–1.5°E, covering 243,438 km² (Figure 1). The country spans diverse ecological gradients, from the humid forests of the south to the dry savannahs in the north. Annual rainfall ranges from 2,000 mm in the forest zones to about 1,000 mm in the Sudanian savannah, with a mean annual temperature near 26 °C (Yamba *et al.,* 2023). Ghana's six agroecological zones i.e. Sudanian, Guinean, Transitional, Rainforest, Semi-deciduous, and Coastal savannahs (figure 1) exhibit wide variation in soils, dominated by Acrisols, Lixisols, Ferralsols, Plinthosols, and Vertisols (Adjei-Gyapong & Asiamah, 2002). These conditions support diverse cropping systems, including cereals and legumes in the north and perennial crops (cocoa, oil palm) in the south.

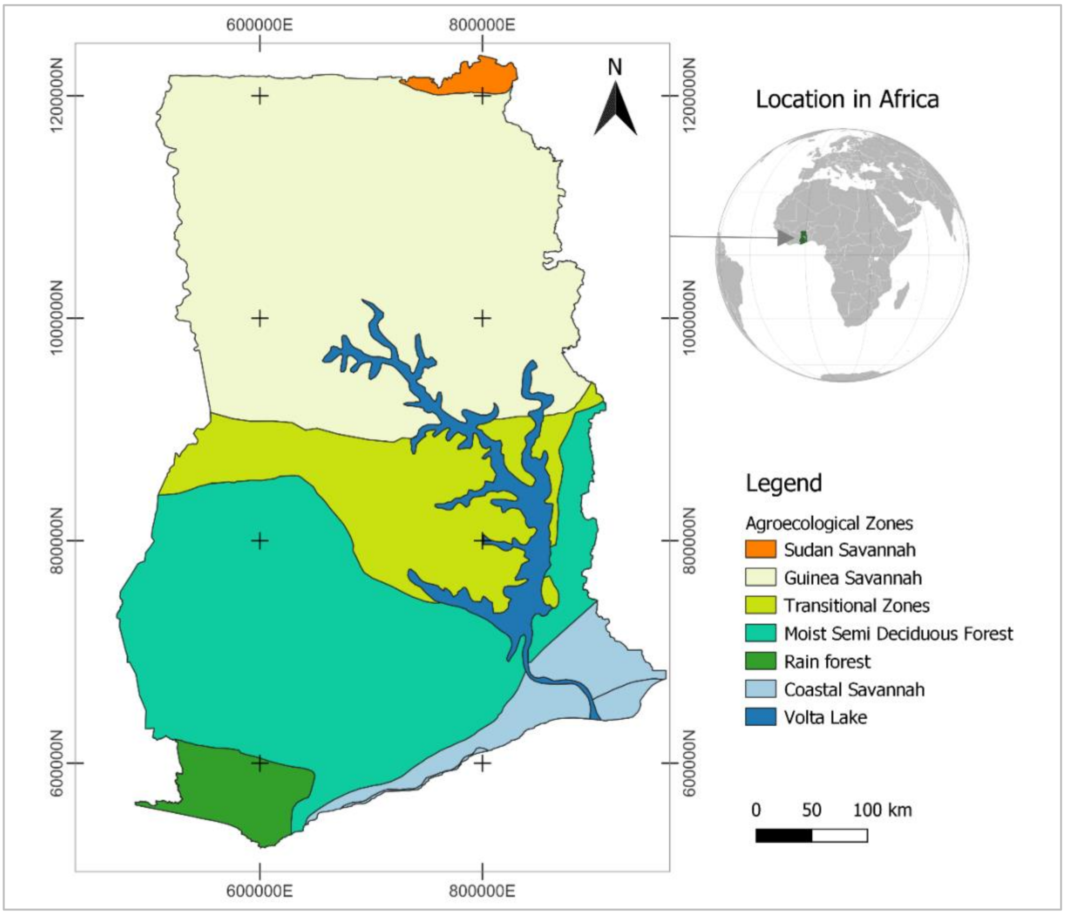

Figure 1 . Map location and agro-climatic zones of Ghana

2.2. Modeling Framework

2.2.1. RothC Model Overview

Soil organic carbon (SOC) dynamics were simulated using the RothC model(Coleman & Jenkinson, 1996), which partitions total SOC into five compartments: decomposable plant material (DPM), resistant plant material (RPM), microbial biomass (BIO), humified organic matter (HUM), and inert organic matter (IOM). Carbon turnover in each pool follows first-order kinetics adjusted by monthly environmental factors- temperature (a), moisture (b), and vegetation (c):

$$\boldsymbol{C_{end} = C_{start} * e^{-abck/t}}$$

With **C:** Amount of organic matter in a specific compartment of the model. **a:** temperature factor; **b**: soil moisture factor; **c**: land cover factor**, k:** annual decomposition constant, and **t:** time expressed in months. Factors **a, b** and **c** adjust the decomposition constant (k), used to represent the decomposition rates of soil organic matter compartments: rapid DPM (decomposable); Slower RPM (resistant); Intermediate BIO (microbial biomass); HUM (humus) very slow and IOM (inert) almost undecomposable (Coleman & Jenkinson, 1996). These decomposition rates allow the analysis of carbon storage in soils, the impacts of agricultural practices and the influence of climate change. The sum of the SOCs in these buckets represents the total active SOC.

2.2.2. Model Inputs and Data Sources

Input variables for the RothC model include initial carbon stock, climate parameters, amount of vegetation cover, soil clay content, and organic carbon inputs (Abera et al., 2025). In the absence of local soil layer, initial soil carbon stocks (t C ha^{-1}, 0–30 cm) and clay content (%) were obtained from ISRIC SoilGrids (https://soilgrids.org) at 250 m resolution. These datasets provided the baseline for RothC initialization. Monthly NDVI data were used to represent cropland biomass dynamics, serving as a proxy for vegetation productivity and residue input to the soil. NDVI was extracted from MODIS MOD13Q1 (250 m, 16-day composites) for 2000–2024 via Google Earth Engine (Didan et al., 2015). Following Huete et al. (2002), a scaling factor of 0.0001 was applied to standardize NDVI values between -1 and 1, aggregated to monthly means. Climate variables (temperature, precipitation, evapotranspiration), which are essential for the RothC model to function, were extracted monthly from various satellite sources via the GEE platform. The average temperature was estimated from the collection MODIS

MOD11A2, which provides the Earth's surface temperature (*Land Surface Temperature*, LST) at a spatial resolution of 1000 meters (Yang et al., 2019). The collection has been filtered according to Ghana's boundaries and over the period 2000 to 2024. Prior to aggregation into monthly averages, the daily LST band was converted from Kelvin to degrees Celsius using the MODIS scaling factor, according to Wan (1999). The formula for conversion is as follows:

$$T_{Celsius} = LST_{brut} \text{ x } 0.02 - 273.15$$

Precipitation data were extracted from CHIRPS (*Climate Hazards Group InfraRed Precipitation with Station data*), which offers near-global coverage (Funk et al., 2015). The data was just aggregated in monthly value over a period from 2000 to 2024.To calculate potential evapotranspiration (in mm/day), the air temperature at 2 meters and the incident solar radiation from the surface were downloaded from the collection ERA5-Land of the European Centre for Medium-Range Weather Forecasts (Muñoz-Sabater et al., 2021). The PET was estimated from the equation of Hargreaves & Samani (1985), adapted to tropical regions was used (Asare et al., 2011).

$$\boldsymbol{ETP = 0.0023 \times \left(T_{moy} + 17.8\right) \times (T_{max} - T_{min})^{0.5} \times RA}$$

Where: Tavg: Average daily temperature (°C), Tmax: Maximum daily temperature (°C); Tmin: Daily minimum temperature (°C); RA**:** Solar radiation (MJ/m²/day).

The analysis was restricted to annual croplands (cereals, legumes, tubers), excluding perennial systems (cocoa, oil palm, etc.). The annual croplands land mask was extracted from the land use database of the Ghana Forestry Commission in 2020 (figure 2). Finally, all datasets were harmonized to 250 m resolution, WGS84/UTM Zone 30N, and national extent.

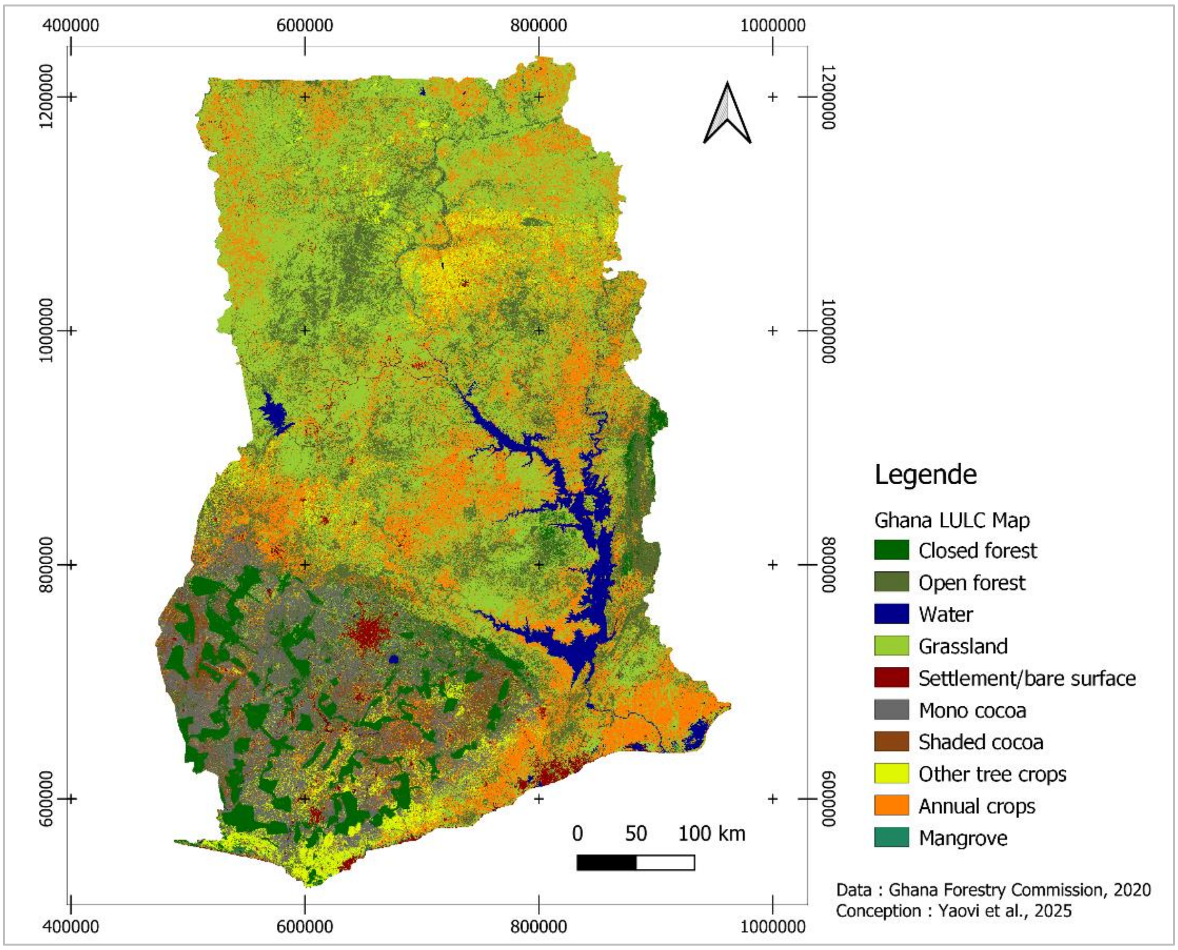

Figure 2 : Land use map of Ghana

2.3. Model Implementation

The RothC model implemented following four steps: (i) initialization of organic matter pools; (ii) calculation of monthly environmental factors; (iii) estimation of monthly carbon fluxes for each pool; and (iv) monthly updating of pools and computation of total SOC stock. Due to limited local fractionation data, pool proportions followed RothC standards (Coleman & Jenkinson, 1996; Falloon et al., 2007): DPM = 3%, RPM = 10%, BIO = 2%, HUM = 70%, IOM = 15%.

The value of the coefficient (a), which is the influence of temperature on the rate of decomposition, was calculated using the following equation:

$$a = \frac{47.91}{1 + \exp\left(\frac{106.06}{T + 18.27}\right)}$$

Where T is the mean monthly temperature. The moisture factor (b) depends on the maximum topsoil moisture deficit (TSMDmax) which is calculated as a function of the clay content before adjusting it to soil depth (Farina et al., 2013). In the standard version of RothC (Coleman & Jenkinson, 1996), it is presented as follows:

$$TSMD_{max} = -(20 + 1.3 \times \%Clay - 0.01 \times \%clay^2) \times \frac{depth}{23}$$

Where ***%Clay*** is the percentage of clay and ***depth*** is the depth of the soil sample which is 30 cm in this study. The moisture deficit accumulated over time (*acc. TSMD*) was calculated from the water balance, comparing the amount of water lost through evapotranspiration to the amount of water provided by precipitation.

$$Balance\ = Precipitation\ -\ PET$$

- If balance < 0: TSMD = TSMD + |Balance|
- Otherwise: TSMD = max (0, TSMD − Balance)

Thus, the calculation of the factor b given as:

$$b = \begin{cases} 0.2 & \boldsymbol{if\ TSMD\ >\ 0.444\ x\ TSMDmax} \\ 0.2\ +\ 0.8\ x\ \dfrac{TSMDmax\ -\ TSMD}{TSMDmax\ -\ 0.444\ x\ TSMDmax} & \boldsymbol{otherwise} \end{cases}$$

The vegetation factor was calculated based on NDVI instead of fixed values. NDVI is correlated with above-ground biomass, which can be used to estimate organic matter inputs to the soil, and makes it possible to capture seasonal variations in plant growth, phenology and senescence (Ayalew et al., 2020). The vegetation factor (c) was calculated using the following equation:

$$c_{factor} = 0.6 + 0.4 \times NDVI_{norm}$$

Were NDVI*norm* is the normalized NDVI calculated using the formula:

$$NDVI_{norm} = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min} + 0.001}$$

Normalized NDVI values were used to estimate monthly carbon inputs (tC/ha/month). With reference to the fixed values proposed by Meersmans et al. (2013) for cropland (1 to 3 tC/ha/year, depending on the type of crop and residue management) and per Farina et al. (2013) for non-cultivated or arid soils (0.1 to 0.5 tC/ha/year, reflecting their low productivity). Carbon inputs (c_{input}) were then calculated as:

$$c_{input} = \frac{NDVI_{norm} \times 0.5 + 0.2}{12}$$

Finaly, the monthly carbon fluxes is calculated based on the simulation of the decomposition of the organic pools and the monthly CO^2 flux. The monthly decomposition rates for each pool were defined as (Skjemstad et al., 2004):

- $\mathrm{decomp}_{DPM} = \mathrm{DPM} \times \frac{10.0}{12} \times a \times b \times c_{factor}$
- $decomp_{RPM} = RPM \times \frac{0.3}{12} \times a \times b \times c_{factor}$
- $decomp_{BIO} = BIO \times \frac{0.66}{12} \times a \times b$
- $decomp_{HUM} = HUM \times \frac{0.02}{12} \times a \times b$

A fraction of the decomposed carbon is lost as CO_2, calculated as:

$$CO_{2_{pool}} = decomp_{pool} \times fCO_{2_{pool}}$$

with

- $fCO2 = 0.84$ for DPM/RPM (high mineralization) and
- $fCO2 = 0.55$ for BIO/HUM (partial stabilization).

The decomposition residues are then redistributed to the BIO and HUM pools with the formulas:

$$\mathrm{B}IO = (decomp_{DPM} + decomp_{RPM}) \times 0.46 \times 0.16 + (decomp_{BIO} + decomp_{HUM}) \times 0.54 \times 0.45$$

$$HUM = \left(decomp_{DPM} + decomp_{RPM}\right) \times 0.16 \times 0.54 + \left(decomp_{BIO} + decomp_{HUM}\right) \times 0.46 \times 0.45$$

Coefficients **0.46** and **0.54** represent the shares allocated to BIO and HUM respectively of these residues, 16% of DPM/RPM flows and 45% of BIO/HUM flows are transferred to the pools (i.e., not lost as CO_2) (Zimmermann et al., 2007).

The carbon pools and stock dynamics are then calculated and updated **monthly** using a mass balance according to the following equations (Coleman et al., 2024):

- $DPM_{t+1} = \max\left(0, DPM_t - decomp_{DPM} + c_{input} \times 0.59\right)$
- $RPM_{t+1} = \max\left(0, RPM_t - decomp_{RPM} + c_{input} \times 0.41\right)$
- $BIO_{t+1} = \max\left(0, BIO_t - decomp_{BIO} + BIO_{flux}\right)$
- $HUM_{t+1} = \max\left(0, HUM_t - decomp_{HUM} + HUM_{flux}\right)$
- $IOM_{t+1} = IOM_t$

2.4. Temporal Analysis and Scenario Assessment

SOC dynamics were analyzed at the pixel level across all croplands in Ghana for the period 2001–2074, covering both historical reconstruction and future projection. The retrospective phase (2001–2024) captures past SOC trends based on observed climatic and vegetation conditions, while the prospective phase (2024–2074) projects future SOC trajectories under alternative climate pathways derived from the IPCC SSP–RCP framework (Riahi et al., 2017; van Vuuren et al., 2011). The three scenarios (Baseline, Optimistic, and Pessimistic) reflect contrasting assumptions on temperature rise, rainfall patterns, and ecosystem response (Table 1).

Table 1 : Description of climate scenarios used for future SOC projections in Ghana (2024–2074)

Scenario	Description
Baseline	Continuation of 2001–2021 conditions (ΔT < 1 °C).
Optimistic (SSP1–2.6)	Low warming (+1 – 1.5 °C), slightly wetter, with improved biomass recovery.
Pessimistic (SSP5–8.5)	High warming (+2.5 – 4.5 °C), more extremes, increased SOC losses.

2.5. Model validation

The performance of the RothC model was evaluated by comparing the simulated values of soil organic carbon (SOC) for cultivated lands in 2024 with those obtained from SoilGrids for the same year. This evaluation was based on several standardized statistical indicators, namely: the Pearson correlation coefficient (r), the coefficient of determination (R^2), the Nash–Sutcliffe efficiency (EF), the root mean square error (RMSE), and the Willmott index (WI).

In order to assess the direct or delayed influence of climatic variables and NDVI on the dynamics of agricultural carbon stocks, we applied regression models with time-lagged variables (lags from 0 to 5 years).

3. RESULTS

3.1.RothC model performance

The initial soil organic carbon (SOC) stock estimated from SoilGrids for Ghana's agricultural lands averaged 35.61 ± 8.50 t C ha^{-1}, whereas the RothC-simulated value was 24.37 ± 5.40 t C ha^{-1}. Model evaluation (Figure 2) showed a strong correlation ($r = 0.91$) between observed and simulated values. However, the Nash–Sutcliffe Efficiency (NSE = –1.16) indicated a systematic underestimation of SOC by the RothC model compared to SoilGrids data. The RMSE (11.51 t C ha^{-1}) and MAE (11.20 t C ha^{-1}) were moderate, and the Willmott Index (WI = 0.62) reflected partial model agreement. Despite the bias, the model effectively captured spatial and relative variation across croplands (Figure 3).

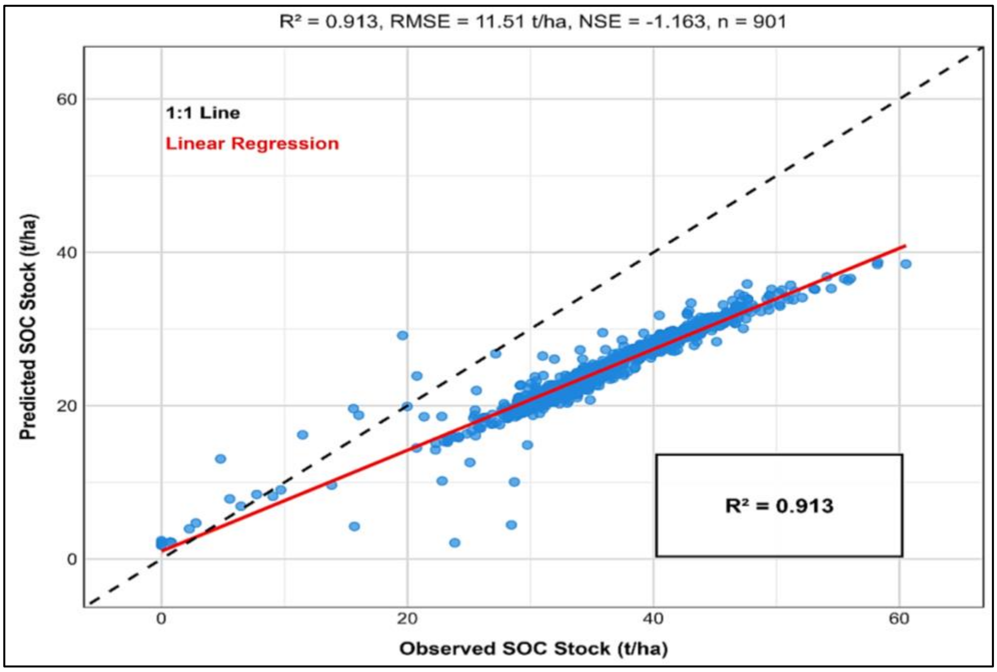

Figure 3 Regression lines observed (*SoilGrids)* and RothC predicted carbon stocks

3.2. Spatio-temporal Dynamics of SOC stock (2001–2024)

SOC stocks exhibited a gradual decline across Ghana's croplands between 2001 and 2024 (Figure 4). The national mean decreased from 34.25 t C ha^{-1} in 2001 to 22.94 t C ha^{-1} in 2024 - an absolute loss of 11.31 t C ha^{-1}, equivalent to 33.0 % reduction.

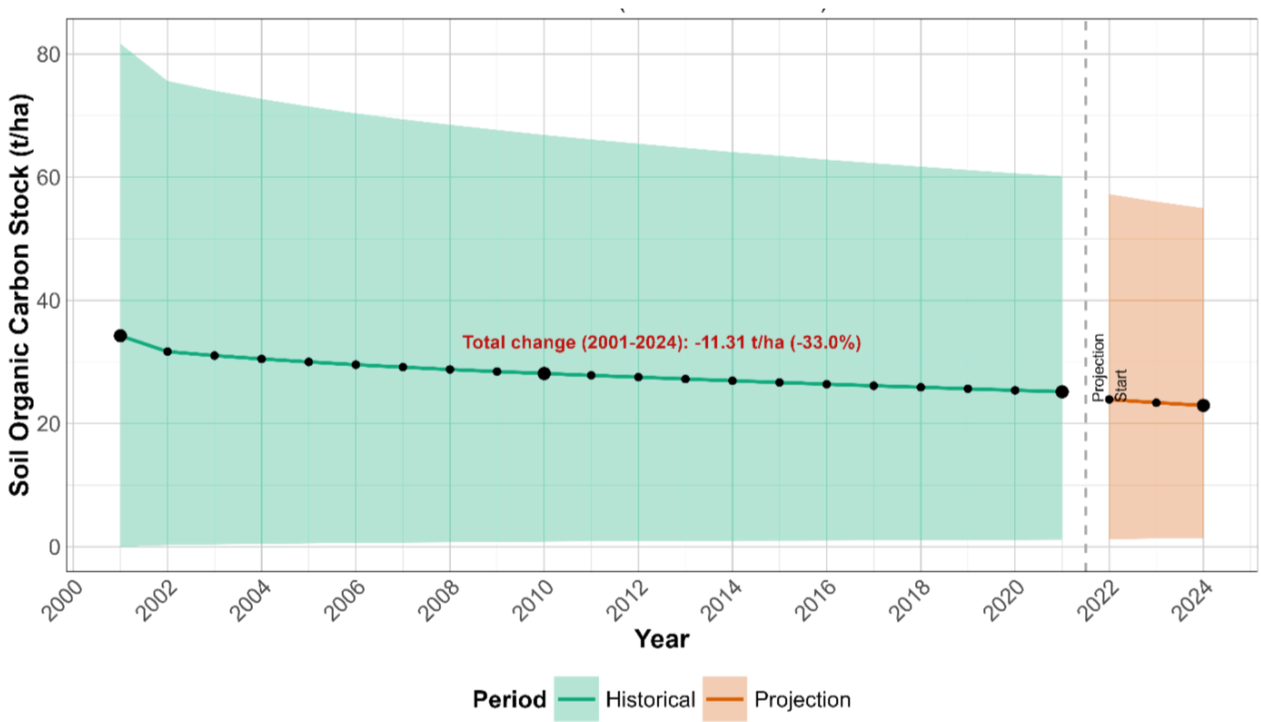

Figure 4 Historical trend in Ghana's farmland carbon stock from 2001 to 2024

The maximum SOC value dropped sharply from 84.9 t C ha^{-1} to 54.9 t C ha^{-1}, whereas the minimum increased slightly from 0.0 to 1.38 t C ha^{-1}, reflecting a narrowing of the SOC range (Figure 4). Spatially, SOC displayed a clear north–south gradient (Figure 5). In 2001, moderate SOC levels (15–30 t C ha^{-1}) dominated central Ghana, while higher concentrations (>30 t C ha^{-1}) were concentrated in the southern forest zones. By 2024, these high-SOC areas had contracted, replaced by moderate and low categories, indicating widespread depletion, especially in the south and central regions (figure 5). The lowest values were found in the north, with a few high-carbon pockets (46.8–62.4 t/ha) along the south-eastern coast. By 2021–2024, SOC had generally declined, with a reduction in high-carbon areas and an expansion of moderate ones, particularly in central regions. Overall, the 2001–2024 change map reveals a widespread decrease in SOC across Ghana.

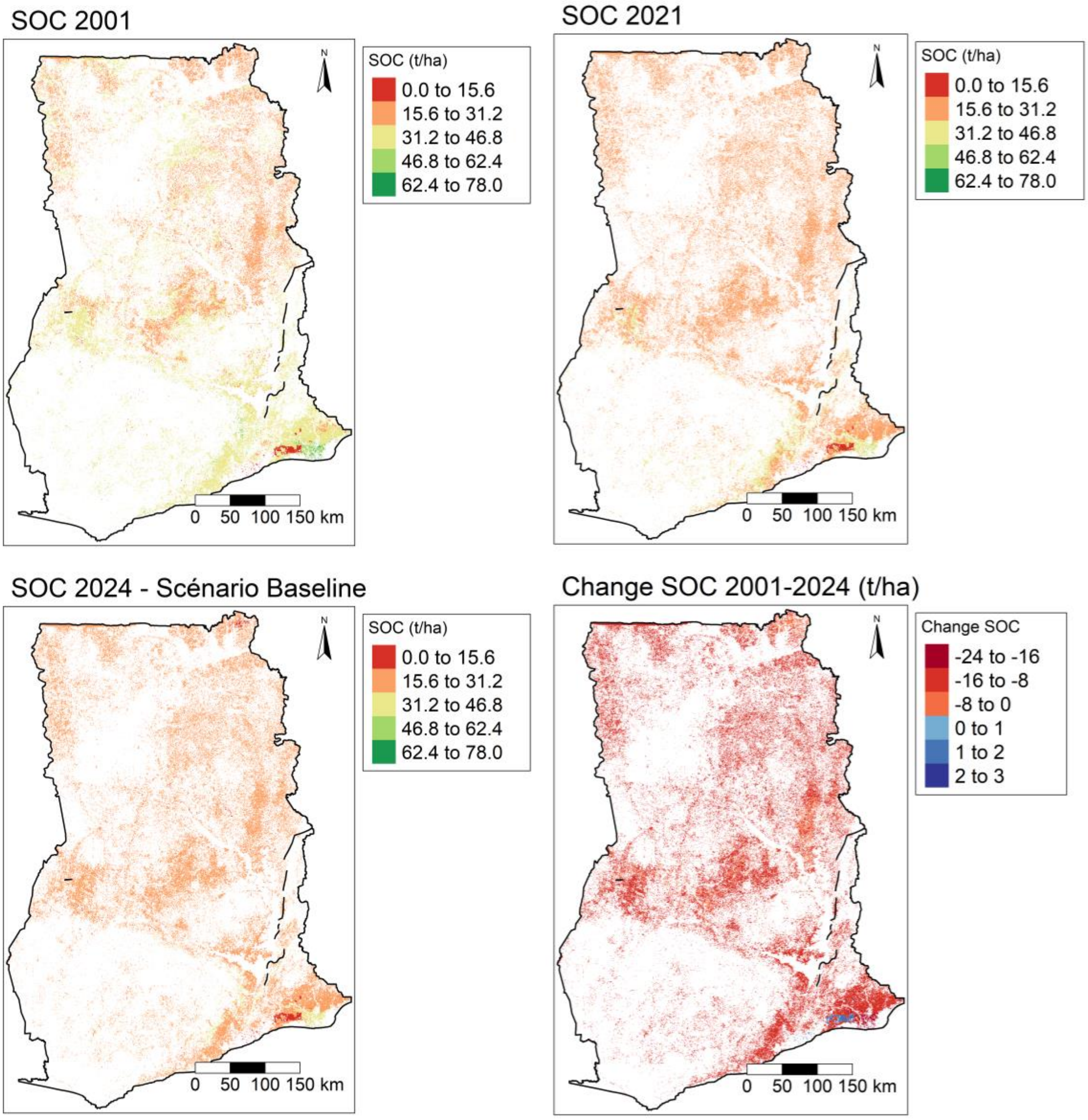

Figure 5 Spatial evolution of SOC stocks in Ghana from 2001 to 2024, as simulated by the RothC model

3.3.Dynamics of SOC stock by agro-ecological zones

SOC stocks declined significantly across all six agroecological zones between 2001 and 2024 (Table 2 ; Figure 6). The Rainforest and Moist Semi-deciduous Forest zones maintained the highest SOC stocks, whereas the Guinea and Sudan Savannas had the lowest. However, all zones exhibited comparable rates of decline -27 to -31% reduction ($p < 0.0001$) - with annual losses ranging from - 0.29 to - 0.39 $tC.ha^{-1} yr^{-1}$. The steepest declines occurred in the humid southern zones, suggesting rapid depletion of historically carbon-rich soils.

Table 2 Temporal trends of SOC stocks (2001–2024) by agroecological zone.

Ecological Zone	SOC 2001 (t/ha)	SOC 2024 (t/ha)	SOC Change (t/ha)	Percentage Change (%)	Annual Trend (t/ha/year)	R^2 ***
Rain forest	38,89	28,34	-10,55	-27,13	-0,388	0.975
Moist Semi Deciduous Forest	37,90	27,45	-10,46	-27,58	-0,384	0.975
Coastal Savannah	35,76	25,17	-10,59	-29,62	-0,391	0.975
Transitional Zones	31,57	22,71	-8,86	-28,07	-0,325	0.976
Guinea Savannah	30,29	20,95	-9,34	-30,84	-0,345	0.977
Sudan Savannah	24,58	16,84	-7,74	-31,48	-0,287	0.976

*** The p-values are all less than 0.0001.

The annual change in SOC for each agro-ecological zone over the 23 years shows a significant decline in all zones (Figure 5). The variations in absolute and relative SOC and the rate of annual loss in Table 1 confirm this and reveal a widespread and statistically significant decline in SOC with very high R^2 (>0.97) and highly significant p-values (< 0.0001). This reflects a continuous and regular trend, with notable variations between zones.

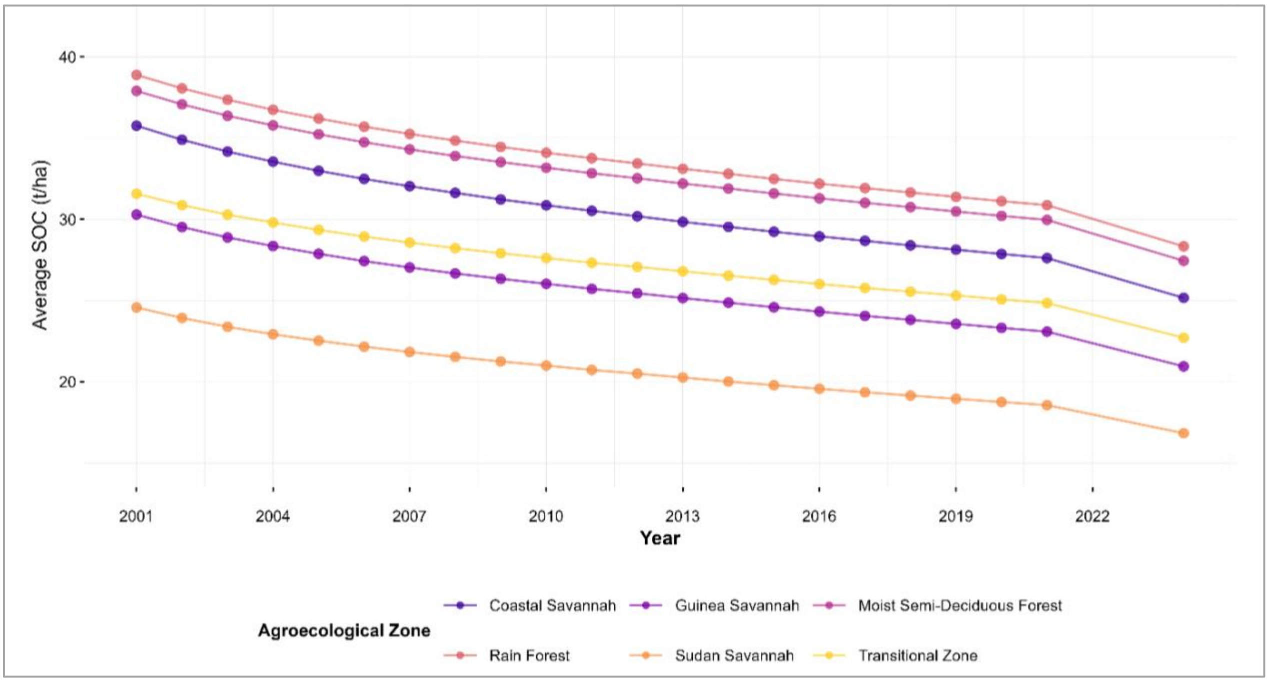

Figure 6: Temporal dynamics of the SOC stock by agro-ecological zones

3.4. Distribution of SOC stock among carbon pools (2001–2024)

The distribution of soil carbon stock among the different pools shows that humus (HUM) is the dominant fraction in 2021, with 22.71 t/ha, or 70% of the total stock, followed by inert organic matter (IOM), the second largest pool with 4.87 t/ha (15%) (figure 7). The fractions of resistant

(RPM) and decomposable (DPM) plant matter are lower, reaching 3.24 t/ha (9.99%) and 0.97 t/ha (2.99%) respectively. Finally, soil microbial biomass (BIO) has the lowest proportion at 0.65 t/ha (2.00%). In 2024, all active basins (HUM, RPM, DPM, BIO) recorded a decrease in their stock, with losses of between -6.6 t/ha and -0.19 t/ha, corresponding to a relatively homogeneous reduction rate of 29% (Figure 7). On the other hand, IOM remained stable (0% change). The highest loss was observed in humus (HUM), which fell to 16.05 t/ha, a decrease of -6.6 t/ha, while the lowest was recorded for microbial biomass, with a reduction of -0.19 t/ha, reaching 0.46 t/ha.

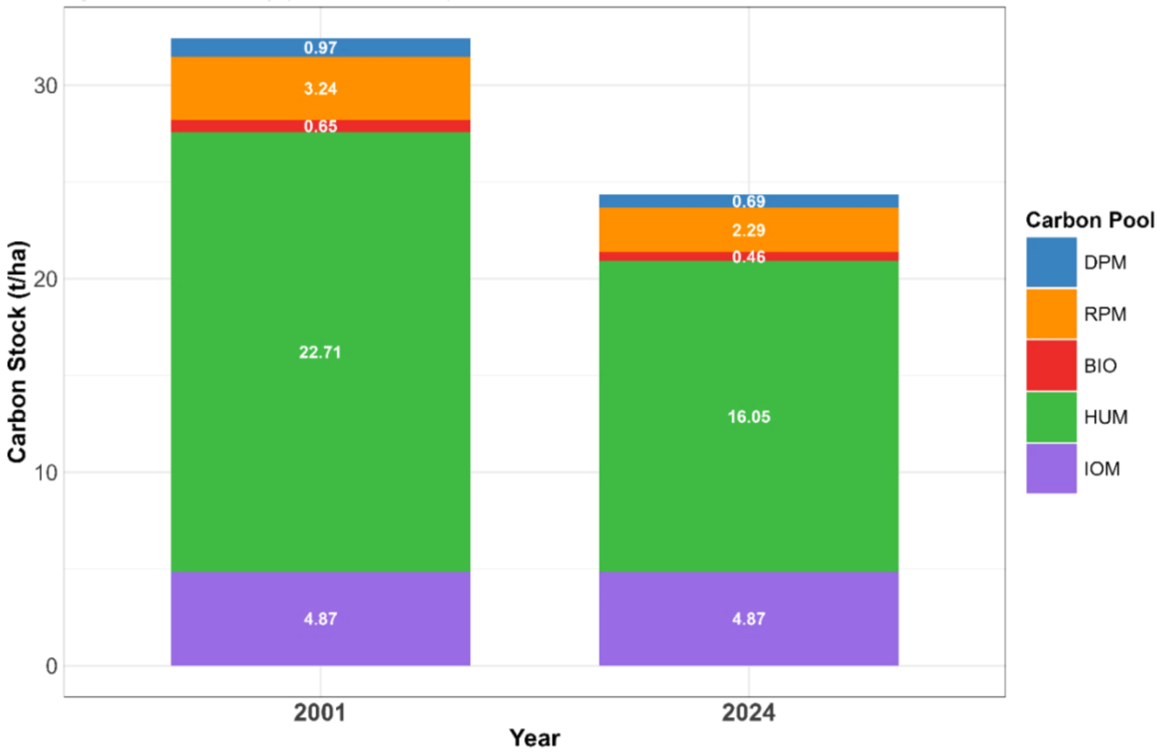

Figure 7 Evolution of the composition of SOC pools between 2001 and 2024

3.5.Direct or lagged effects of climate variables and NDVI on SOC stock dynamics

The table 3 shows the regression coefficients between climatic factors and carbon stock in Ghana's agricultural land with time-shifted variables over 2 years. The results indicate that between 2001 and 2021, the water balance does not have an immediate effect on the SOC stock, but its impact becomes negative and almost significant after 2 years (-0.44, pvalue = 0.06), suggesting that a water deficit gradually reduces organic matter inputs. Similarly, potential evapotranspiration (PET) has no immediate effect, but after 2 years, its influence becomes positive (0.44, pvalue = 0.06), suggesting that it promotes the decomposition of organic matter in the long term. The vegetation index (NDVI) has a moderate negative correlation with the SOC, with no significant delayed effects, which could be explained by a rapid decomposition of plant residues. Finally, temperature has a nearly significant, moderate negative direct effect (-0.38, pvalue= 0.09), which decreases over time, indicating that high temperatures initially accelerate the decomposition of the SOC before stabilization of the more resistant fractions.

Table 3 : Regression model coefficients with time-shifted variables

Factor	Lag (year)	Correlation	p_value
Water balance	0	0,03	0.89
Water balance	1	-0,24	0.30
Water balance	**2**	**-0,44**	**0.06**
PET	0	-0,03	0.89
PET	1	0,24	0.30
PET	**2**	**0,44**	**0.06**
Precipitation	0	-0,04	0.86
Precipitation	1	0,04	0.87
Precipitation	2	0,02	0.95
Temperature	**0**	**-0,38**	**0.09**
Temperature	1	-0,35	0.13
Temperature	2	-0,29	0.22
NDVI	0	-0,28	0.22
NDVI	1	-0,16	0.51
NDVI	2	-0,14	0.55

3.6.Future SOC Trajectories Under Climate Scenarios (2021–2074)

The RothC simulations indicate that Ghana's agricultural SOC stocks will continue to decline under all climate scenarios (Figure 8). From an estimated 24.7 $tC.ha^{-1}$ in 2021, SOC is projected to fall to 20.3 $tC.ha^{-1}$ under the Optimistic scenario, 18.0 $tC.ha^{-1}$ under the Baseline, and 15.7 $tC.ha^{-1}$ under the Pessimistic scenario by 2074 (Figure 8).

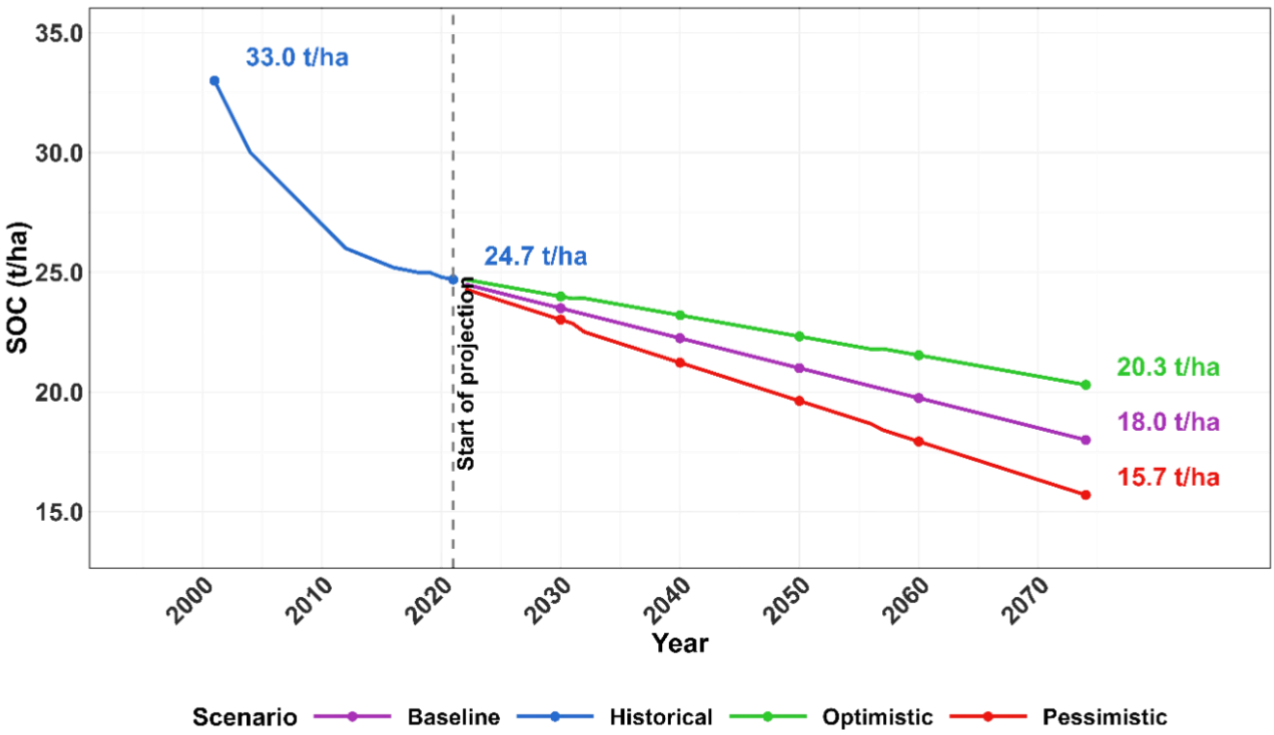

Figure 8 Projected SOC stock trends (2021–2074) under three climate scenarios

Spatial projections (Figure 10) show persistent north–south contrasts, with the lowest SOC (<19 $tC.ha^{-1}$) in northern croplands and the highest in southern forested areas. However, all scenarios exhibit widespread declines, particularly in the south. Under the Baseline scenario, southern SOC losses reach –30 to –22 $tC.ha^{-1}$, while the Optimistic scenario mitigates declines to –22 to –15 $tC.ha^{-1}$. The Pessimistic scenario results in extensive depletion across the country, with large areas falling below 10 $tC.ha^{-1}$ by 2074.

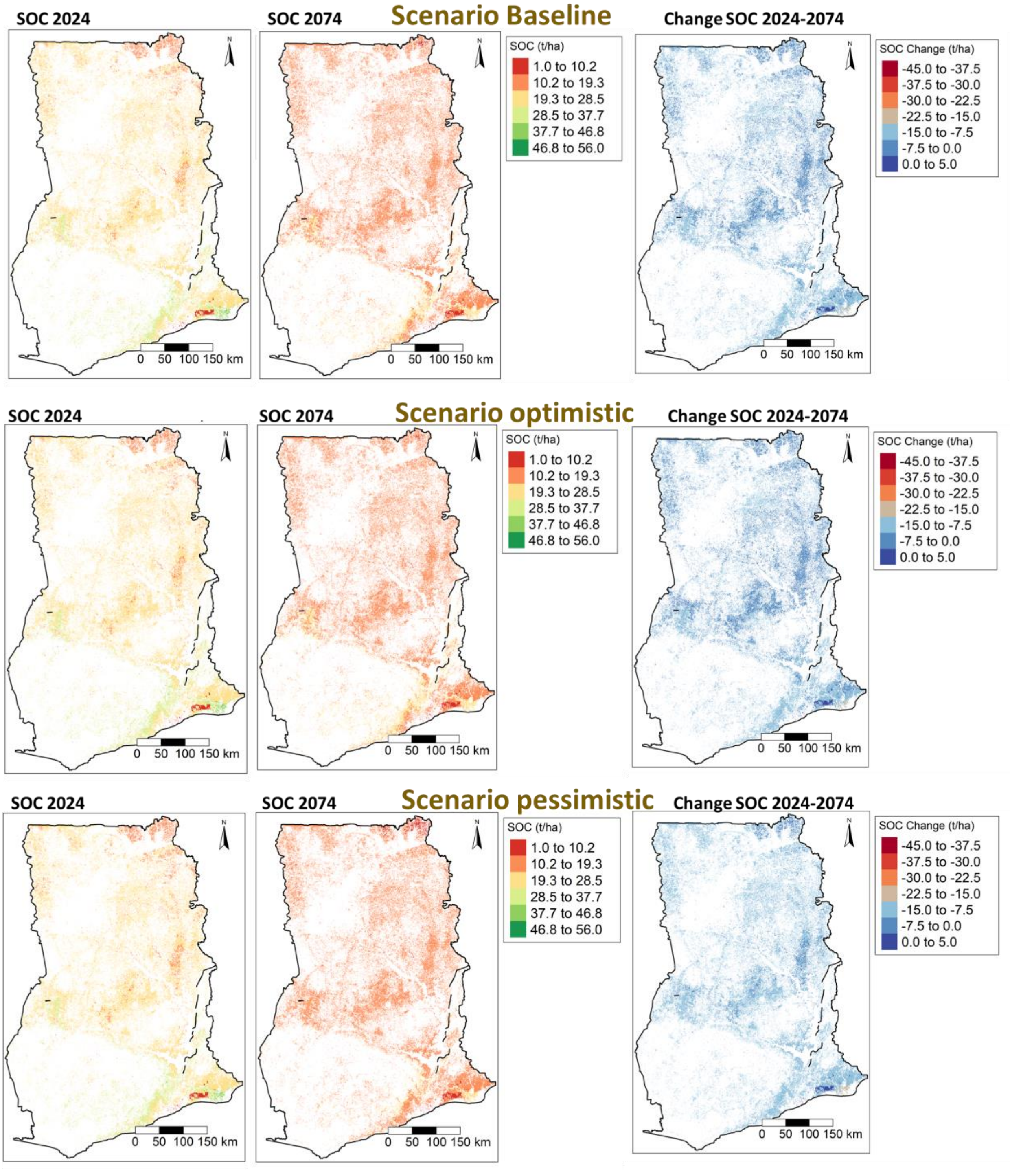

Figure 9 Spatial distribution and percentage change in SOC stock (2024 vs. 2074) under Baseline, Optimistic, and Pessimistic scenarios

4. DISCUSSION

Performance of the RothC model

This study aims to improve understanding of the organic carbon dynamics of agricultural soils in Ghana over the last two decades and into the future, through simulation of carbon stocks using the RothC model.

The results show discrepancy between the observed carbon stock (SoilGrids: 35.61 ± 8.50 tC/ha) and the stock simulated by the RothC model (24.37 ± 5.40 tC/ha) for Ghana's agricultural land. The performance of the RothC model is mixed because, although it has a strong correlation (r = 0.91) and an acceptable Willmott index (d = 0.62), the negative NSE (-1.16) and moderate errors (RMSE = 11.51 tC/ha; MAE = 1.20 tC/ha) indicate a systematic underestimation bias. However, RothC's predictions align with values reported by previous studies of agricultural areas in Ghana, thus 12 to 33 tC/ha (or average ~22.5 tC/ha) in the Guinean savannah and transitional zones (Bessah et al., 2016) or 7.58 to 25.56 tC/ha depending on the crops grown (maize, rice, etc.) in the district of Tolon in the Northern region (Nyadzi et al., 2019). The observed discrepancy could therefore be explained by an overestimation of the SOC in SoilGrids. Indeed, the average SOC of SoilGrids appears to be overestimated compared to the stocks of SOC reported for West African agricultural land, generally less than 30 tC/ha (Anokye et al., 2021; Vågen et al., 2005). This discrepancy suggests that the SoilGrids layer captures an inherited soil state that includes the influence of forest ecosystems where SOCs can reach 60 to 100 tC/ha (Amankwah et al., 2024; Zhang-Zheng et al., 2024). Hengl et al. (2017) mentioned that the first version of SoilGrids (at 1 km resolution) tended to overestimate low soil organic carbon (SOC) values. Therefore, if RothC is biased towards SoilGrids, it likely provides values closer to the agricultural reality of Ghana as well as also in West African agricultural systems where the SOC is between 20 and 30 tC/ha (Bationo et al., 2007; Zougmoré et al., 2010).

Spatio-temporal SOC trends (2001–2024)

The historical evolution of the soil organic carbon (SOC) stock in Ghana's agricultural land over the last two decades (2001 to 2024) shows a general trend of gradual decrease in the average stock with a loss of 0.85 t/ha (or 3.6%). This trend is confirmed by several scientific studies conducted in Ghana, which highlight a depletion of SOC related to changes in land use, conversion of forests to cropland, intensive agricultural practices such as conventional ploughing and climate change (Amankwah et al., 2024; Owusu et al., 2020).

The present study reveals a significant decline in the maximum initial value of the SOC (84.9 t/ha to 54.9 t/ha), indicating a marked degradation in areas with high initial potential. This decrease of more than 30 t/ha is mainly the result of the conversion of forests to cropland and intensive agricultural practices (Anokye et al., 2021; Owusu et al., 2020). The work of Anokye et al. (2021) confirms this, as they showed that agricultural land had significantly lower stocks than neighbouring forests with a similar deviation of 35 t/ha (73.5 vs. 108.2 Mg/ha). Naab et al. (2017) report rapid decreases of 24 to 38% in SOC over four years after conversion of native vegetation to cropland in northwestern Ghana, attributed to the loss of organic matter due to tillage.

The observed increase in the minimum SOC (+1.38 t/ha) indicates an improvement in initially poor soils. This trend is likely related to the adoption of sustainable management practices and climate-smart technologies promoted through institutional projects , resulting in positive effects in areas with low SOC potential despite the overall decrease observed (Dziwornu et al., 2025). Dziwornu et al. (2025) listed 153 climate-smart agriculture (CSA) projects implemented between 1971 and 2023, mostly concentrated in the Northern and Upper West and Upper East regions where the SOC potential is much low and threatened. These efforts confirm the spatio-temporal heterogeneity of the SOC observed in the present study, with losses being more marked in areas initially rich in SOC (up to −30 t/ha), while some poor soils recorded modest gains (+1.38 t/ha). Owusu et al. (2020) confirm this spatial variability of the SOC with systematically lower stocks in the North (0.05–43.5 Mg/ha over 0–30 cm) and losses intensifying from the southwest to the northeast. This confirms results of this study on the dynamics of the SOC according to the agroecological zones of Ghana.

SOC dynamics across agro-ecological zones and carbon pools

Soil organic carbon (SOC) stocks follow a decreasing gradient from moist forests to dry savannahs: Rain forest > Moist semi-deciduous forest > Coastal savannah > Transitional zone > Guinea savannah > Sudan savannah, reflecting Ghana's south-north bioclimatic gradient. This order is consistent with ecological observations, with forests storing more carbon thanks to abundant biomass and slower mineralization, while dry savannahs show lower stocks due to accelerated decomposition (Bessah et al., 2016; Brown et al., 2024; Owusu et al., 2020).

Between 2001 and 2024, all zones record a significant drop in SOC ($p < 0.0001$), with absolute losses of -7.74 to -10.59 t/ha and relative losses of -27 to -31%. Even forest ecosystems, initially rich in carbon, show marked vulnerability, while the Coastal savannah records the highest absolute loss (-10.59 t/ha), probably linked to strong agricultural and urban pressure (Anokye et al., 2021; Naab et al., 2017).

The examination of carbon pools between 2001 and 2024 shows that humus remains the dominant fraction of the SOC (≈70%), but is also the most vulnerable, accounting for the largest losses (–6.6 t/ha). This result corroborates those of Geremew et al. (2024) in Ethiopia and Dechow et al. (2019) in Europe, underlining the sensitivity of the HUM pool to changes in land use and agricultural intensification. Active pools (DPM, RPM, BIO) recorded a proportional reduction of about 29%, reflecting a degradation of labile and intermediate organic matter. This trend corroborates with simulations of Pesce et al. (2024), who reported a high level of responsiveness of labile pools to management practices and organic inputs.

Impact of climate variables on SOC dynamics in agricultural land

The results show how climatic factors contribute to accelerating or undelaying the degradation of SOC in a context of climate change in Ghana. Overall, increasing temperatures and water variations accelates carbon loss. This corroborates the works of Soares et al. (2023) and Zhao et al. (2021). This study also reveals a delayed effect of water deficit and ETP on the SOC, suggesting that a water deficit gradually reduces organic matter inputs and SOC replenishment. On the other hand, temperature has a direct negative effect, thus confirming its immediate threat to labile carbon stocks. This observation is in line with the observations of several authors (Hartley et al., 2021; Sun et al., 2023; Wang et al., 2022), but contrasts with the results of Woo & Seo (2022), who showed that a significant decrease in SOC only occurs after a delay of about four years.

The moderate negative effect of NDVI on SOC in Ghana's croplands (farmlands), while surprising, could be explained by the low contribution of annual crops to organic carbon accumulation. Similar results have been reported by Zhang et al. (2019), which found negative correlations between NDVI and SOC in January, April and December in China's Jianghan Plain.

Moreover, these complex temporal dynamics highlight the increased vulnerability of agricultural soils to the loss of SOCs under the combined effect of water deficit and global warming that characterizes sub-Sahelian Africa. In the case of Ghana, this supports the idea that the observed decline in SOC is not only due to changes in land use or agricultural practices, but also to immediate or delayed interactions of soil temperature and moisture.

Projected SOC trajectories under climate scenarios (2021–2074)

RothC model projections predict a continued degradation of soil organic carbon (SOC) stocks in Ghanaian agricultural systems over the period 2021-2074. The extent of this decline is strongly conditioned by management practices, with a moderate decline (–17%) under an optimistic scenario, a substantial loss (–25.8%) in the baseline scenario, and a pronounced deterioration (–34.5%) in a pessimistic scenario. These trajectories confirm the substantial risk of a persistent depletion of the SOC if nothing is done, with an annual rate of loss ranging from –0.08 to –0.17 t/ha/year. The results are consistent with those of Gottschalk et al. (2012) and

Loum et al. (2014) indicating that optimistic management scenarios, such as agroforestry, organic amendments and reduced tillage, are likely to significantly slow down or even reverse the loss of SOC.

CONCLUSION

This study, using RothC model, reveals a widespread decrease in soil organic carbon (SOC) stocks in Ghana's agricultural land between 2001 and 2024, with higher losses in areas with high initial potential (up to –30 t/ha) and modest gains in poor soils (+1.38 t/ha). Although RothC underestimates stocks compared to SoilGrids (24.37 vs 35.61 tC/ha), its estimates are more consistent with the values observed in Ghana and West African farming systems (20–30 tC/ha). The losses, concentrated in the humus pool, are mainly the result of intensive agricultural practices, deforestation, and climatic constraints (water deficit, global warming). 50-year projections indicate a continued degradation of the SOC (–17 to –34.5%), making climate change a critical limiting factor for carbon sequestration. These findings underscore the urgency of adopting agroecological and climate-smart practices and integrating mitigation into agricultural policies to strengthen the sustainability of Ghana's production systems.

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